CS781 Capstone Project: In-House Legal Intelligence System

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Public repository: https://github.com/anshumanr24/CS781-Capstone

Abstract

Indian legal text is privileged, so the stack has to be trained on the official CS-781 corpus and served on premises. This project is one family of four from-scratch language models (50M–300M parameters, shared byte-level BPE, 1024-token block) and seven task heads built on copies of those weights. The design question is how to reuse a compact decoder as a token encoder: rotary positions and grouped-query attention for pretraining, LoRMA and related PEFT for transfer, character-aligned windows for NER, sentence-level CRFs for roles, multi-window pooling for long judgments, and lexical first-stage retrieval where a 256-token neural view is too short. Official ranks and primary metrics are in Table 1; the body of the report describes the architectures that produced them.

1 Competition Result

Codabench username: a_240148 **Roll number:** 240148

Table 1 is the selected zip of each phase, with rank as shown on the official board (freeze 9 November 2026). At least four distinct systems were uploaded per phase (four Phase-1 architectures, LoRMA, a second PEFT method, and a distillation student).

Table 1: Official selected zip per phase. Rank is the live-board rank of that zip on 9 Nov 2026.

Phase	Task	Primary metric	Rank	Distinct zips
1	Legal LM pretraining	BERTScore F1 0.8328	55	$\geq$4
2	Legal NER	strict macro-F1 0.5446	51	$\geq$4
3	Rhetorical roles	macro-F1 0.5290	39	$\geq$4
4	Court judgment prediction	macro-F1 0.6763	96	$\geq$4
5	Statute identification	macro-F1 0.4252	23	$\geq$4
6	Summarization	BERTScore F1 0.8503	22	$\geq$4
7	Prior-case retrieval	Micro-F1@7 0.4519	33	$\geq$4

Total valid Codabench submissions across all phases: $\geq$ 35. **Date ranks were recorded:** 9 November 2026.

2 Introduction and Problem Description

A firm that holds Indian judgments cannot send them to a hosted API or initialise from a public legal BERT. The capstone therefore requires an in-house pipeline: one foundation family trained only on `Exploration-Lab/CS-781-Capstone` [1], then seven heads. Phase 1 continues a legal prompt for up to 128 words. Phase 2 extracts twelve entity types with exact character offsets. Phase 3 labels every provided sentence with one of thirteen rhetorical roles. Phase 4 predicts a binary outcome. Phase 5 assigns a subset of 100 masked statute IDs. Phase 6 writes a summary. Phase 7 returns seven prior cases from a split-specific pool.

The constraints that shaped the architecture were a 50M–300M parameter cap, no external corpora, a Tesla T4 memory budget, and strict span matching for NER. I did not train a single multi-task net and did not pipe Phase-2/3 outputs into later phases, because an offset error would propagate. Public checkpoints and LLM APIs were not used.

3 Data Analysis and Preprocessing

3.1 Sources and official splits

All JSON comes from the official Hugging Face configuration. Table 2 is counted from the files actually loaded.

Table 2: Official split sizes.

Phase	Unit	Train	Dev	Test
1	continuation prompts (test)	SC JSON pretrain	—	1000
2	documents (3 folds)	35+35	8 held out of fold 2	35
3	documents (CL / IT)	40 / 40	5 / 5	5 / 5
4	cases	37 387	1 988	1 517
5	documents, 100 labels	42 750	10 181	13 019
6	judgment / summary pairs	7 030	—	100
7	queries / candidates	827 / 4 320	118 / 1 023	237 / 1 727

3.2 Corpus statistics that informed design

Phase 1 pretraining is dominated by Supreme Court prose; high-court and tribunal dumps are longer-tailed and more formulaic, so the selected efficient run stayed on SC text inside a 1024-token block. Byte-level BPE with a word-end marker keeps citation tokens (“AIR”, years, “SCC”) as a few merges.

Phase 2 has twelve types. APP, RESP, COURT and STAT are common; WIT and R.COUNSEL are rare. Spans are short but can sit after several thousand characters of letterhead, so a word-limit that stops at 400 words never sees them. Adjacent APP/RESP mentions occur; nested spans are rare.

Phase 3 shares thirteen roles across corporate-law and IT. Fact, Argument and Ratio dominate; Dissent and PrecedentOverruled are almost absent, which is why a document-level CRF with a rare-role bias is part of the head rather than a second encoder. Sentences are the official lists.

Phase 4 is binary. The operative sentence usually sits near the end of the judgment, so a head-only 512-token crop deletes the signal. Phase 5 has a long tail of statute IDs (about 3.7 labels per document) and the *mentions* of those statutes are masked in the text. Phase 6 gold summaries

overlap heavily with source n-grams and are much shorter than the judgments, which argues for sentence selection rather than free generation. Phase 7 queries and candidates are the same long, name-masked documents; query IDs that also appear in the pool must be dropped.

3.3 Preprocessing

The same BPE is used everywhere. Spaces and case are kept so character offsets invert. Each BPE token stores `[start, end)` in the original UTF-8 string. Phase 2 slides overlapping 1024-token windows and merges duplicate spans. Phase 3 does not re-segment. Phases 4–6 pack a document as a few windows (head+tail or evenly spread). Phase 7 clips to 8 000 words as two-thirds head plus one-third tail, matching how a reader skims a judgment.

4 Approach and Model Evolution

4.1 Phase-1 foundation model

Four architectures share a tied input/output embedding and a 1024-token block. None uses `nn.Transformer`. `build_model` refuses a count outside $[50, 300]$ million.

Table 3: The four Phase-1 bodies. All share vocab 8192, block 1024, tied embeddings, and a `build_model` check in $[50, 300]$M.

Name	Attention / positions	MLP / norm	Params
decoder	full causal MHA, learned pos.	GELU, LayerNorm	110.2M
hybrid	256-token window on 3/4 layers	same as decoder	110.2M
efficient	RoPE, GQA 16Q/4KV	SwiGLU, RMSNorm	118.5M
encdec	6 bi-enc. + 6 causal+cross	GELU, LayerNorm	106.3M

The **decoder** is a pre-norm causal Transformer [2]: width 1024, 8 layers, 16 heads. It is the control architecture—standard MHA with no rotary bias—so later ablations can be attributed to packing or PEFT rather than to a new positional scheme.

The **hybrid** keeps that size, but three of every four layers mask attention to a 256-token causal band. Local legal phrasing ("it is hereby ordered", a statute short-title) then costs $O(n \cdot w)$ rather than $O(n^2)$. Every fourth layer is full causal attention so a citation at token 40 can still reach token 900. Residual connections and the FFN are unchanged, which is why the parameter count matches the decoder.

The **encoder–decoder** uses width 768, six bidirectional encoder blocks and six causal decoder blocks with cross-attention. A training block of 1024 tokens is split in the middle: the first half is the encoder source, the second half is teacher-forced on the decoder. That split is *not* used at task time; Phase 6 reuses the same weights with a full document as source and a summary as target.

The **efficient** model (width 1024, 10 layers) is the body copied into most later phases. Rotary positions [3] replace a learned table so relative distance is in the attention logits. RMSNorm replaces LayerNorm. The FFN is SwiGLU [4] with an expansion that keeps the count near 118M. Grouped-query attention (16 query heads, 4 KV heads) [5] shrinks the KV cache on T4 without dropping to multi-query.

The tokenizer is a custom BPE [6]. A pair is merged only if it occurs at least twice; the score is Dice-weighted, $2C(xy)^2/(C(x)+C(y))$, so merges prefer cohesive legal pieces rather than fragments of "the". Bytes 0–255 are always in the vocab. Training is next-token cross-entropy, AdamW, block

1024, on Supreme Court text. Decoding is greedy and stops at EOS instead of padding to a fixed word count.

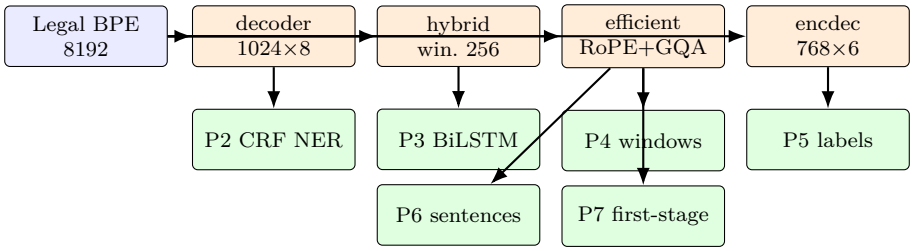


Figure 1: One BPE, four Phase-1 backbones, seven separately adapted descendants. No multi-task trunk.

4.2 Transfer protocol

Every later phase loads a Phase-1 `state_dict` and the same BPE. Adaptation is LoRMA [7]

$$h = \big(I + (\alpha/r)\, BA\big)Wx, \tag{1}$$

with B initialised at zero so the first step matches the frozen LM; LoRA [8]; Houlsby adapters [9]; partial unfreezing of the last k blocks; BitFit; and response distillation [10] from the efficient teacher into an encoder–decoder student. LoRMA is run on at least one architecture in every task phase. Full fine-tuning of the efficient body is used only for NER, where the T4 still fits.

4.3 Phase 2 — Legal NER

The head is token classification on BPE ids, then reconstruction of character spans from stored offsets—not a word BiLSTM that would re-tokenise the line. Each token carries a BIO tag; a linear-chain CRF [11] on that sequence forbids I-X after B-Y. Decoding is Viterbi. Because a judgment is longer than 1024 tokens, overlapping windows are encoded independently and duplicate spans keep the higher-scoring copy. A light post-process attaches APP to an adjacent person-like mention and strips punctuation. The selected system averages two efficient full-FT+CRF seeds. The required PEFT set is efficient LoRMA, decoder full FT, hybrid LoRA, and encdec distillation; distillation failed when the student did not inherit the offset tables.

4.4 Phase 3 — Rhetorical roles

Each official sentence is encoded by the Phase-1 model (mean pool of non-pad hidden states). A BiLSTM over that sentence sequence, then a CRF, predicts one of thirteen roles, so left/right context is the rest of the judgment rather than extra tokens inside the sentence encoder. The sentence encoder stays frozen under LoRMA; only the low-rank factors, the BiLSTM and the CRF are trained at full rate. Corporate-law and IT documents share the role set and are trained jointly. A rarity bias on Dissent and PrecedentOverruled is part of the CRF potential. The other required PEFT methods are decoder IA^3 and hybrid LoRA; the student is encdec distillation from the efficient LoRMA teacher.

4.5 Phase 4 — Court judgment prediction

A case is packed as several 256-token windows spread from head to tail, with extra mass on the ending. Hidden states are mean-pooled, a two-way linear head produces a logit, and the cut is

chosen on dev macro-F1 (accuracy is the wrong target on a two-class problem). The last ≈100 words are also encoded as a bag-of-words "ending cue" and concatenated to the neural logit, because the holding is usually written there. Partial FT unfreezes only the last transformer blocks of the efficient body so the T4 can keep a batch of four windows. Hybrid adapters and decoder BitFit are the other PEFT submissions; encdec is the distillation student.

4.6 Phase 5 — Statute identification

Up to six 256-token windows are encoded. A per-label attention head over tokens is added to a mean-pooled linear path whose bias is the label-frequency prior: each of the 100 IDs has its own query vector rather than a shared 100-way softmax. Loss is binary cross-entropy. Empty predictions are replaced by the label with the largest score/threshold ratio. Thresholds are fit per label on a two-fold split of dev, because a single cut of 0.5 cannot serve both the head and the tail of the frequency histogram. Independently, a TF-IDF + per-label logistic regression is trained on the same corpus (statute strings are masked, so this is not a leak of section numbers). The selected system blends the neural logits with the linear model. The four-arch PEFT set is efficient LoRMA, decoder partial, hybrid adapters, encdec distillation.

4.7 Phase 6 — Summarization

Causal models are trained as document + BOS + summary with loss only on summary tokens. The encoder–decoder encodes the judgment and decodes the summary (its Phase-1 `forward` split is *not* reused). That abstractive path is fluent but free to invent a holding. The selected system instead ranks source sentences from encoder–decoder hidden states and emits an extractive summary with a 720-word budget—length control is the budget, not a coverage penalty. Sentence scores are a linear map of the pooled hidden state; greedy selection adds a sentence if it still fits the budget. LoRMA, partial FT, adapters and KD were all run on the abstractive path to meet the PEFT requirement.

4.8 Phase 7 — Prior-case retrieval

The first stage is a split-specific TF-IDF index (word n-grams 1–3, 400k features, sublinear TF) over an 8k-word clip of each document. Cosine scores the query against the pool, the query's own ID is removed, and seven IDs are returned. A small citation-span field and a 1200-word head field are z-scored and added with small coefficients; they do not replace the first stage. Dual-encoder InfoNCE on 4×256 windows and a Phase-1 pair (cross) encoder that reorders only the TF-IDF top 20–30 were implemented for the neural/PEFT requirement. They read too little of a name-masked judgment to override a lexical list, so they remain ablation architectures rather than the serving path.

5 Experiments and Implementation Details

5.1 Compute and training protocol

Neural jobs ran on Kaggle Tesla T4 (16 GB, internet off, two concurrent kernels per account). Checkpoints write `best.pt` on the local metric; a `--train_minutes` cap kills the job. Optimizer is AdamW (weight decay 0.01), gradient clip 1.0. Dropout 0.1 sits only on newly added heads. Early stopping tracks span quality (P2), role macro-F1 (P3), thresholded macro-F1 (P4/P5), summary overlap (P6), and retrieval F1 on the 118 labelled queries (P7).

Table 4: Hyperparameters of the selected architecture in each phase.

	P1	P2	P3	P4	P5	P6	P7
Body	eff. GQA	eff. full	eff. LoRMA	eff. partial	blend	encdec rank	TF-IDF
Packing	1024 blk	1024 slide	sent pool	6×256	6×256	sentences	8k words
Peak LR	1e−4	2e−5	1e−4	2e−5	1e−4	—	—
Batch	8	2	1 doc	4	2	—	—
Schedule	≈116M tok.	5–8 ep.	8 ep.	6–8 ep.	4 ep.+LR	extract	fit/split
Decode	greedy+EOS	Viterbi	Viterbi	dev τ	per-label τ	720w	K=7

5.2 Local evaluation

Held-out rows are always a subset of public train/dev. The official scripts under `evaluators/` are run before every upload (schema, unique IDs, offsets, exactly seven PCR IDs). Local and hidden test diverge when sklearn versions differ, when a ROUGE combo is used as a proxy for BERTScore, or when labelled dev is folded into training.

5.3 Ablations

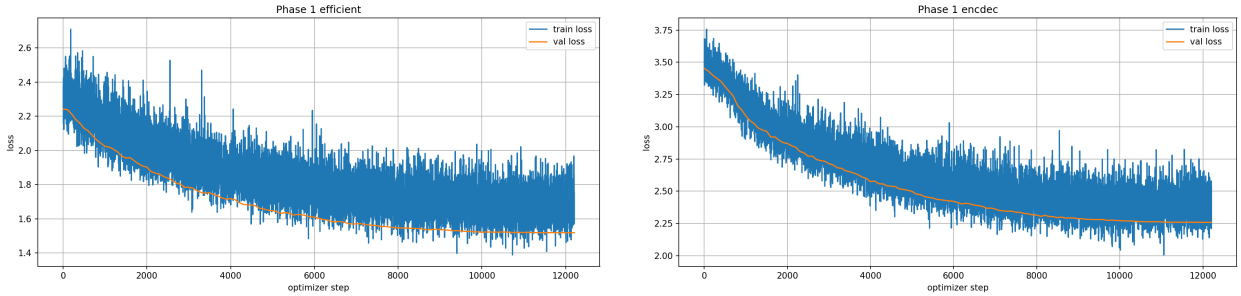
One-factor changes: Dice BPE versus count BPE; head+tail packing versus a cropped head; per-label versus global statute thresholds; extractive sentence rank versus abstractive continuation; TF-IDF first stage versus a dual-encoder or pair encoder on the same backbone; LoRMA versus full FT on NER. Negative ablations (pair rerank that overwrites the lexical list; 12k-word PCR that overfits 118 queries; a 400-word NER cutoff) are kept in the code tree.

6 Results

Table 1 is the official record. Table 5 names the architecture of the selected zip and of one runner-up, without repeating the leaderboard. The efficient GQA decoder is the default neural body. Where the task is matching long, masked documents, a linear model on the same corpus is the serving path and the Phase-1 net is a PEFT/KD vehicle.

Table 5: Selected architecture and a runner-up. Numbers live in Table 1.

Phase	Selected architecture	Runner-up
1	efficient, greedy+EOS	decoder / hybrid / encdec
2	efficient full-FT + CRF, 2 seeds	efficient LoRMA+CRF
3	efficient LoRMA + BiLSTM-CRF	hybrid LoRA / decoder IA^3
4	efficient partial + ending cue	efficient LoRMA
5	TF-IDF $\oplus$ efficient windows	TF-IDF alone; PEFT-only heads
6	encdec extractive, 720-word budget	abstractive LoRMA / KD
7	8k TF-IDF + cite/head fields	dual-encoder / pair-encoder



(a) Efficient (selected body)

(b) Encoder–decoder

Figure 2: Phase-1 train and validation loss on T4 (≈12k steps). Validation tracks training; these two bodies are the ones reused downstream.

7 Error Analysis and Discussion

7.1 Phase-wise failure modes

Phase 1 copies the prompt when EOS is late and invents locally fluent citations. Phase 2 misses by a comma or a court suffix; strict span matching treats that as a full miss, so typing quality is understated. Phase 3 confuses Fact with Argument and cannot predict Dissent on documents that simply lack one; PrecedentReliedUpon versus PrecedentOverruled needs more discourse than a mean-pooled sentence. Phase 4 fails on a holding buried in a mid-document double negative, and on "appeal dismissed" quoted from a lower court. Phase 5 misses rare IDs and predicts a frequent neighbour of a masked section. Phase 6 extractive summaries drop a short holding below the word budget; abstractive summaries invent one. Phase 7 leaves relevant cases outside the seven-slot list when they share little surface form with the query (names are masked).

7.2 What the metrics do not measure

ROUGE-L and BERTScore reward overlap with a hidden reference, not legal correctness of a holding [12, 13]. Strict span-F1 rewards offset perfection. Micro-F1@7 treats a fifth relevant case beyond $K{=}7$ as a miss. A deployed stack would still need a lawyer on the output.

7.3 Limitations and next experiments

The 1024-token block and the ban on extra corpora are the ceiling. Under the spec, the next architectural move is a pair scorer that *adds* a feature to TF-IDF rather than replacing the list, and a Phase-6 selector that uses Phase-3 roles as a skeleton. I would not pipeline NER into retrieval until offsets are audited.

7.4 Surprising findings

A from-scratch 118M model is a usable token encoder (NER, roles, judgment windows) and a weak semantic matcher on 8k-word name-masked text. That is why extractive selection beat abstractive continuation, and why a lexical first stage remains the PCR serving path even after LoRMA dual-encoders and pair encoders were trained.

8 Conclusion

The system is four in-house legal LMs and seven separately adapted heads. The architectural choices that mattered were the efficient GQA body, character-accurate BPE offsets, window packing for long judgments, per-label heads for 100 statutes, extractive length control, and a lexical first stage for retrieval. The choices that did not were a cropped-head encoder, abstractive summarization at this size, and a neural ranker that overwrites TF-IDF. A compact legal LM under a parameter and data lock is worth keeping as an encoder; it is not, by itself, a retrieval model.

LLM Usage Declaration

Completing this section is mandatory. Inconsistent answers relative to the code and the prose will be treated as an integrity issue. **Q1.** Which Large Language Model(s), if any, did you use while completing the competition or writing this report? Give model name and vendor (e.g., none / ChatGPT-4.x / Gemini x / Claude x / Grok / Copilot / other).

Answer: Cursor Grok 4.6 (xAI model hosted in Cursor) for coding assistance and for typesetting this report. No ChatGPT, Claude, Gemini, or public LLM API was used to train or to score models. **Q2.** For each row, mark Yes or No and state the purpose in one short phrase. "Yes" is acceptable only for the uses permitted by the capstone document.

Part	Used LLM?	Purpose of using the LLM
Understanding concepts / papers	Yes	Clarify PEFT / CRF / BERTScore definitions
Ideation / brainstorming at a high level	Yes	Enumerate legal ablations already suggested by the spec
Debugging small ancillary snippets	Yes	Sklearn `np.matrix` errors, Kaggle unpack, zip schema
Plotting / visualization code	Yes	TikZ system diagram in this PDF
Grammar / writing style of the report	Yes	Tighten prose against the template checklist
Report structure / section wording	Yes	Map experiment logs onto the required sections
Designing the model architecture	No	Four backbones and LoRMA/CRF heads are student-designed
Hyperparameter search / tuning	No	Windows, LRs, word caps chosen from student runs
Analysis of results / conclusions	No	Failure modes taken from own logs
Writing the core training / inference code	No	Trainers written to the student's spec; LLM used as editor
Other (specify)	Yes	Kaggle kernel packaging; GitHub README typesetting

Q3. Did you refer to sources other than lecture slides and an LLM? If yes, list them (papers, blogs, GitHub repositories, textbooks, official starter notebook, evaluation scripts). These must also appear in the bibliography when they influenced the method.

Answer: Yes. Official CS-781 starter and evaluators; Hugging Face card [1]; papers in the bibliography; sklearn TF-IDF; Kaggle notebook API. **Q4.** Apart from any permitted LLM use, what

was *your* contribution? In 8–12 sentences describe your own understanding, architecture decisions, implementation, experiments, and analysis. Name at least two things you tried that failed and what you concluded from them.

I specified the four Phase-1 architectures (causal decoder, windowed hybrid, GQA-efficient, encoder–decoder), the Dice-weighted BPE, and the rule that every later phase must load those weights. I chose LoRMA as the compulsory PEFT method, CRF heads for sequence tasks, window packing for long judgments, and per-label thresholds for 100 statutes. I ran the Codabench loop: local evaluator, zip, upload, and the decision to keep or drop a zip. I rejected public legal BERT once the spec forbade it. Two failures that changed the design: a 400-word NER BiLSTM that never saw late entities, which forced character-aligned sliding windows; and dual-encoder / pair-encoder PCR that overwrote a lexical list it could not improve. A third failure, a 12k-word PCR clip, overfit 118 dev queries. Extractive 720-word selection was my decision after abstractive SFT stalled. The conclusions in this report are those architectural outcomes.

References

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